

Name: Thabo Aphone

Student number: 57709408

Module: SME 3701

Assignment: 02

Question 1:

1.1

$$k_t = \frac{GJ_p}{L}$$

$$\therefore J_p = \frac{\pi d^4}{32}$$

$$= \frac{\pi(0.050)^4}{32}$$

$$= 6.136 \times 10^{-7} \text{ m}^4$$

$$\therefore k_t = \frac{(79.3 \times 10^9)(6.136 \times 10^{-7})}{1}$$

$$= 486578.70 \text{ Nm/rad}$$

$$J_o \ddot{\theta} + K_t \theta = M_t(t)$$

$$\therefore T = \frac{60}{1000} = 0.06 \text{ s}$$

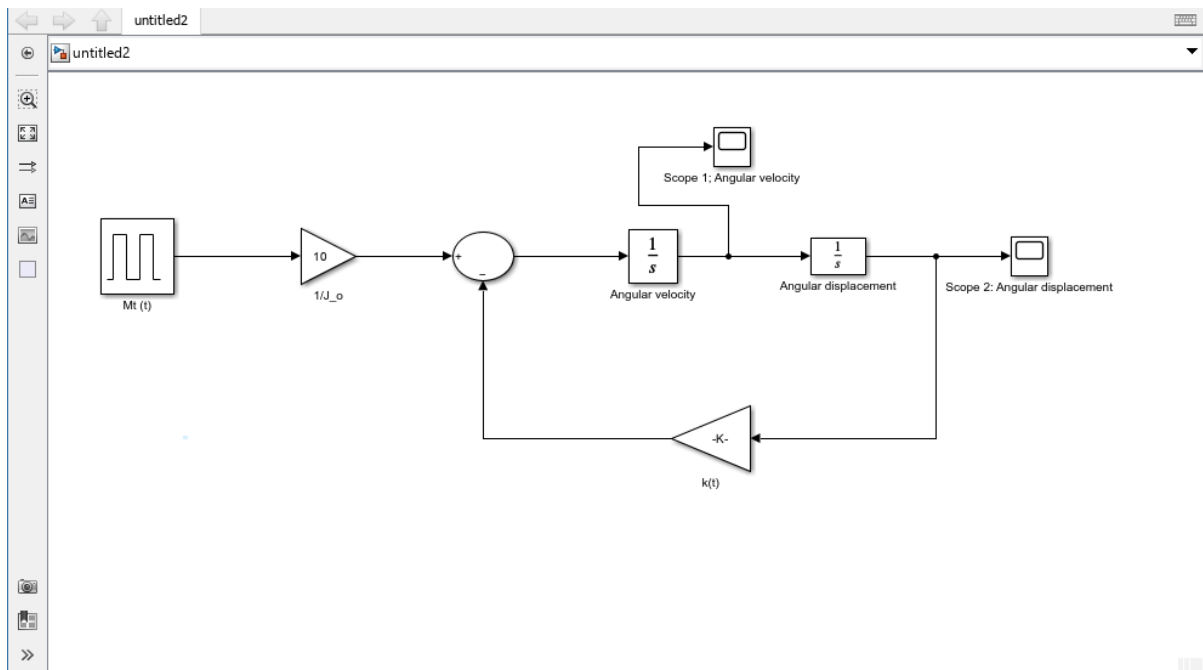
$$\ddot{\theta} = \frac{1}{J_o} M_t(t) - \frac{K_t \theta}{J_o}$$

$$= 10 M_t(t) - 48657870 \theta$$

$$\therefore \text{Pulse width: } \frac{1 \text{ broken tooth}}{16 \text{ total teeth}} \times 100 = 6.25\%$$

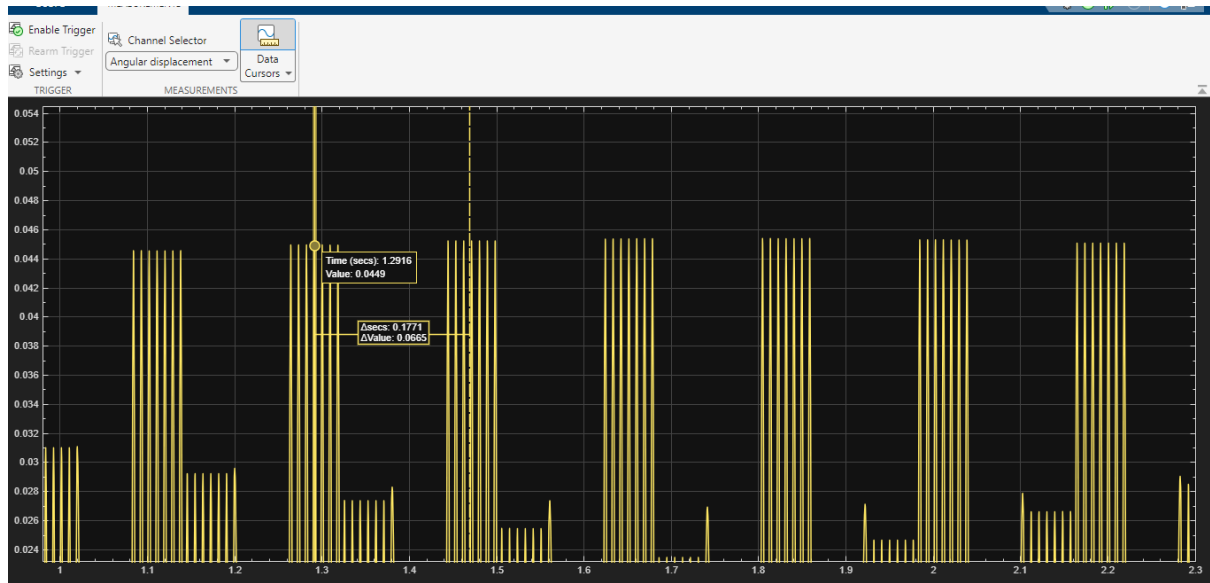
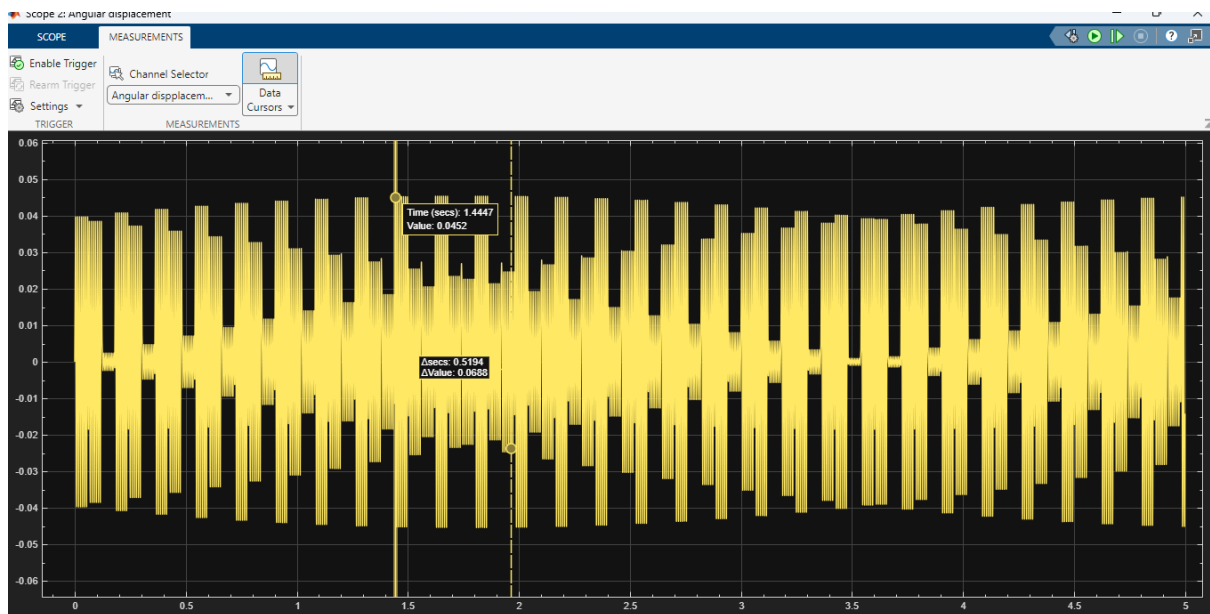
Parameter	Value	Unit
Driven gear inertia J_o	0.10	kg.m^2
Shaft diameter d	0.050	m
Shaft length L	1.0	m
Shear modulus G	79.3	GPa
Driving gear speed	1000	rpm
Transmitted torque	1000	N.m
Number of teeth	16	-
Broken teeth	1	-
Pulse width	6.25	%

Simulink model of the geared shaft system

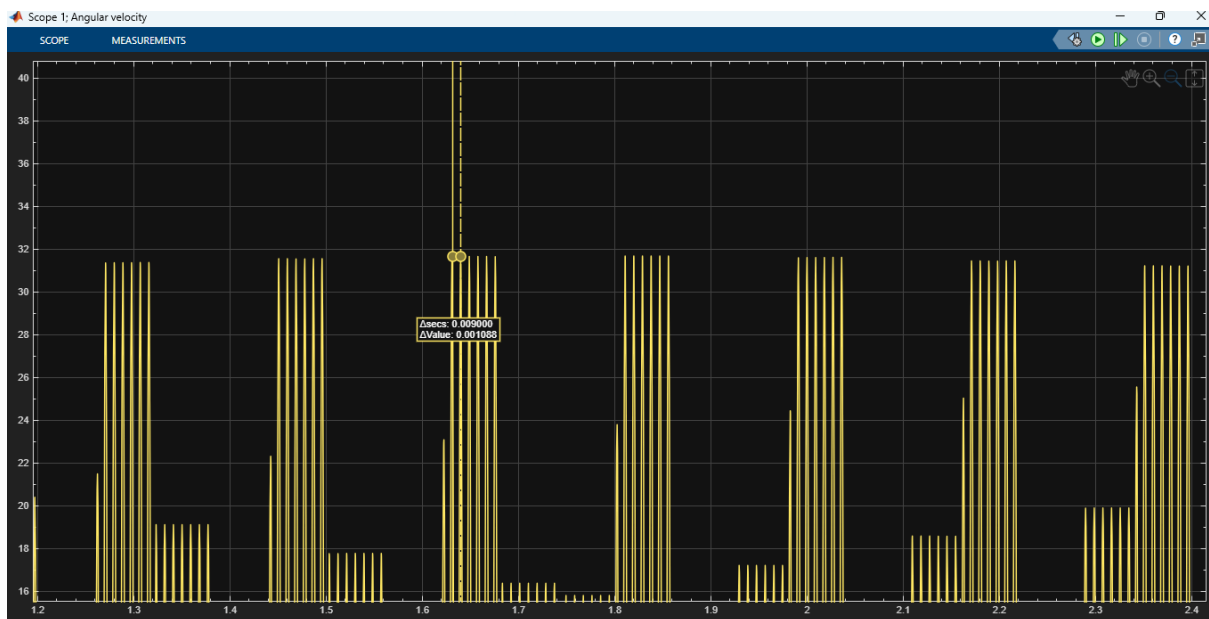
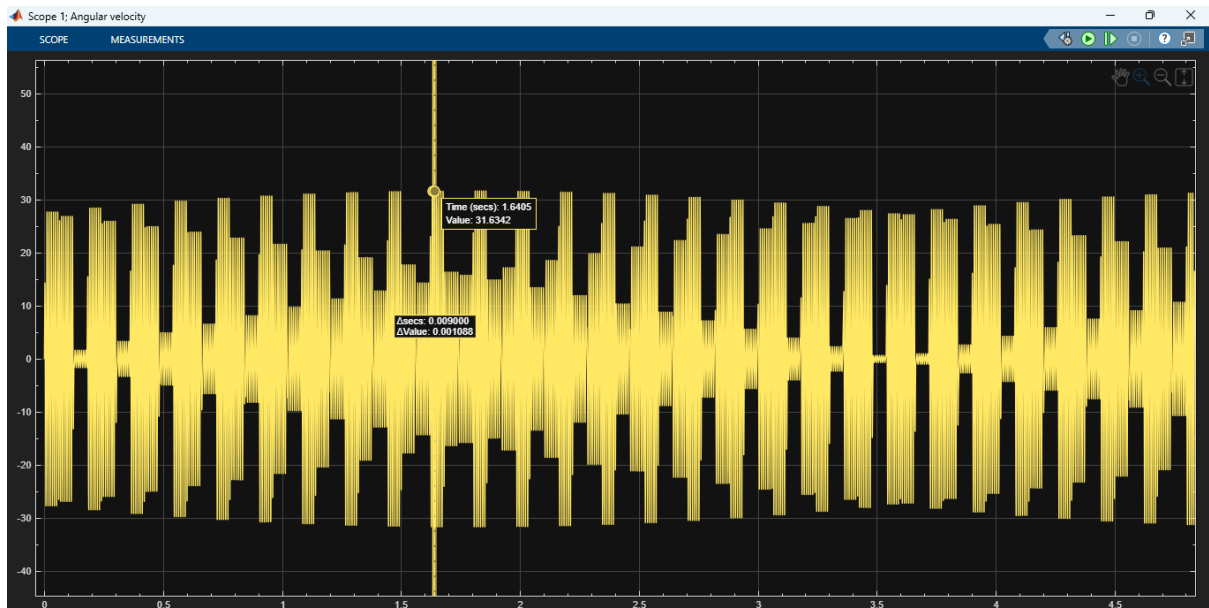


1.2

Angular Displacement:



Angular velocity:



1.3

From the Simulink simulations:

- Maximum angular displacement = 0.0049 rad
- Maximum angular velocity = 31.6353 rad/s

1.4

$$T = 0.00900 \text{ s}$$

∴ The dominant torsional vibration frequency of the driven gear is:

$$\begin{aligned} f &= \frac{1}{T} \\ &= \frac{1}{0.00900} \\ f &= 111.11 \text{ Hz} \end{aligned}$$

Because there are 16 teeth, a broken tooth means the torque drops and recovers once per revolution.

The response is oscillatory with repeated disturbances, which is what we expect from the periodic broken-tooth excitation.

The response shows sustained periodic oscillations rather than a decaying transient response. This indicates that the system is experiencing resonant or near-resonant torsional vibration, caused by the periodic torque disturbance associated with the broken gear tooth.

Simulink Recording:

<https://drive.google.com/file/d/1tk71JUvMr3ixfeioWcslNEg8fIIF99ac/view?usp=sharing>

GitHub repository:

<https://github.com/ThaboAphane/Geard-shaft-torsional-vobration-SME3701-57709408>